



26-12-2025_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT Jee-Main_GTM-05_Test Syllabus

MATHEMATICS : Theory Based Test on TOTAL SYLLABUS

PHYSICS : Theory Based Test on TOTAL SYLLABUS

CHEMISTRY : Theory Based Test on TOTAL SYLLABUS

Sec: Sr. XL-500, Sr.S60(N&S) & LIIT

Page 2



Sri Chaitanya
Educational Institutions



**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

**JEE MAIN
2023**

**SINGARAJU
VENKAT KOURDINYA**
AIR 1
SRI CHAITANYA
SR. XL-500 CLASS

RANK

1

**JEE Advanced
2023**

**NAVILALA
CHIVILAS REDDY**
AIR 1
SRI CHAITANYA
SR. XL-500 CLASS

RANK

1

**NEET
2023**

**BORA VARUN
CHAKRAVARTI**
AIR 1
SRI CHAITANYA
SR. XL-500 CLASS

RANK

1

For More Join: @ITTJEE_Advanced

**MATHEMATICS****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- STATEMENT-1: For $f(x) = \{x\}$, x is a point of minima $\forall x \in \mathbb{Z}$ where $\{.\}$ is fractional part of x .

STATEMENT-2: Let a function be defined as $f(x) = \sum_{n=1}^{10} n|x-n|$, then the minimum value of the function occurs at $x=6$.

 - Statement-1 is True, statement-2 is True
 - Statement-1 is False, Statement-2 is False
 - Statement-1 is True, Statement-2 is False.
 - Statement-1 is False, Statement-2 is True.
- If the incident ray on a surface in XY-plane is along the unit vector $\hat{v}$, the reflected ray is along the unit vector $\hat{w}$ and the normal in the plane of incidence and reflection is along the unit vector $\hat{a}$ outwards where the acute angle between the incident ray and the normal is $\frac{\pi}{5}$ then

 - $\hat{w} = \hat{v} + \left(\frac{\sqrt{5}-1}{2}\right)\hat{a}$
 - $\hat{w} = \hat{v} + \frac{\sqrt{10+2\sqrt{5}}}{2}\hat{a}$
 - $\hat{w} = \hat{v} + \left(\frac{\sqrt{5}+1}{2}\right)\hat{a}$
 - $\hat{w} = \hat{v} + \frac{\sqrt{10-2\sqrt{5}}}{2}\hat{a}$
- STATEMENT-1: Let $F(x)$ be an indefinite integral of $\sin^2 x$ w.r.t x . The function $F(x)$ satisfies $F(x+\pi) = F(x)$ for all real x .

STATEMENT-2: If $\int_a^b f(x)dx = 0$ and $f(x)$ is continuous on $[a, b]$, then the equation $f(x) = 0$ has at least one root in the interval $[a, b]$

Sec: **Sr. XL-500, Sr.S60(N&S) & LIIT**

Page 3



Sri Chaitanya
Educational Institutions



**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

**JEE MAIN
2023**

**SINGARAJU
VENKAT KONDINNA**

ALL-INDIA RANK 1

341

RANK

1

**JEE Advanced
2023**

**VAVILALA
CHIVILAS REDDY**

ALL-INDIA RANK 1

341

RANK

1

**NEET
2023**

**BORA VARUN
CHAKRAVARTI**

ALL-INDIA RANK 1

22

RANK

1

For More Join: @ITTJEE_Advanced



- 1) Statement-1 is True, statement-2 is True
- 2) Statement-1 is False, Statement-2 is False
- 3) Statement-1 is True, Statement-2 is False.
- 4) Statement-1 is False, Statement-2 is True.

4. STATEMENT-1: Let E, F and G be three events having probabilities

$$P(E) = \frac{1}{8}, P(F) = \frac{1}{6} \text{ and } P(G) = \frac{1}{4}, \text{ and let } P(E \cap F \cap G) = \frac{1}{10}.$$

For any event H , if H^C denotes its complement, then $P(E^C \cap F^C \cap G^C) \leq \frac{5}{12}$

STATEMENT-2: Let $S = \{w_1, w_2, \dots\}$ be the sample space associated with a random experiment.

Let $P(w_n) = \frac{P(w_{n-1})}{2}, n \geq 2$. $A = \{2k+3l; k, l \in N\}$ and $B = \{w_n; n \in A\}$ Then $P(B)$ is equal to $\frac{1}{16}$

- 1) Statement-1 is True, statement-2 is True
- 2) Statement-1 is False, Statement-2 is False
- 3) Statement-1 is True, Statement-2 is False.
- 4) Statement-1 is False, Statement-2 is True.

5. Let $\theta_1, \theta_2, \dots, \theta_{10}$ be positive valued angles (in radian) such that $\theta_1 + \theta_2 + \dots + \theta_{10} = 2\pi$.

Define the complex numbers $z_1 = e^{i\theta_1}, z_k = z_{k-1}e^{i\theta_k}$ for $k = 2, 3, \dots, 10$, where $i = \sqrt{-1}$.

Consider the statements P and Q given below:

$$P: |z_2 - z_1| + |z_3 - z_2| + \dots + |z_{10} - z_9| + |z_1 - z_{10}| \leq 2\pi,$$

$$Q: |z_2^2 - z_1^2| + |z_3^2 - z_2^2| + \dots + |z_{10}^2 - z_9^2| + |z_1^2 - z_{10}^2| \leq 4\pi$$

- 1) P is TRUE and Q is FALSE
- 2) Q is TRUE and P is FALSE
- 3) both P and Q are TRUE
- 4) both P and Q are FALSE



JEE MAIN

2023

SINGARAJU

VERMA KORDINIA

RANK 1

341

RANK

1

JEE Advanced

2023

VAVILALA

CHIVILAS REDDY

RANK 1

341

RANK

1

NEET

2023

BORA VARUN

CHAKRAVARTI

RANK 1

722

RANK

1



6. STATEMENT-1: $\lim_{x \rightarrow \infty} \frac{|\sin x|}{\sqrt{1 - \cos^2 x}}$ is neither prime nor a composite number

STATEMENT-2: $\lim_{n \rightarrow \infty} \left\{ \left(3^{\frac{1}{2}} - 3^{\frac{1}{3}} \right) \left(3^{\frac{1}{2}} - 3^{\frac{1}{5}} \right) \dots \left(3^{\frac{1}{2}} - 3^{\frac{1}{2n+1}} \right) \right\}$ is equal to 0

- 1) Statement-1 is True, statement-2 is True
- 2) Statement-1 is False, Statement-2 is False
- 3) Statement-1 is True, Statement-2 is False.
- 4) Statement-1 is False, Statement-2 is True.

7. STATEMENT-1: The function, $f(x) = x \sin\left(\frac{1}{x}\right) \sin\left(\frac{1}{x \sin\left(\frac{1}{x}\right)}\right)$ if $x \neq 0$ and $f(0) = 0$ is

discontinuous at $x = 0$ and the function $g(x) = x \tan\left(\frac{1}{x}\right)$ is discontinuous at $x = 0$

STATEMENT-2: If both $f(x)$ and $g(x)$ are differentiable functions on $\mathbb{R}$, then the function defined as $h(x) = \max\{f(x), g(x)\}$ is not always differentiable on $\mathbb{R}$.

- 1) Statement-1 is True, statement-2 is True
- 2) Statement-1 is False, Statement-2 is False
- 3) Statement-1 is True, Statement-2 is False.
- 4) Statement-1 is False, Statement-2 is true.

8. STATEMENT-1: Let A and B be two 2×2 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the systems of linear equations

$(A^2 B^2 - B^2 A^2)X = O$ (where X is a 2×1 column matrix of unknown variables and O is a 2×1 null matrix) has non trivial solutions only.

STATEMENT-2: Let a, b, c be three non-zero real numbers in which no two of them are equal. Then following system of equations in x, y and z ($x, y, z \in \mathbb{R}$)

$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1, \frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, -\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ has either unique or infinite solutions





- 1) Statement-1 is True, statement-2 is True
 2) Statement-1 is False, Statement-2 is False
 3) Statement-1 is True, Statement-2 is False.
 4) Statement-1 is False, Statement-2 is True.
9. If n is a positive integer, then $(\sqrt{3} + 1)^{2026} - (\sqrt{3} - 1)^{2026}$ is
- 1) an irrational number
 2) an odd positive integer
 3) an even positive integer.
 4) a rational number other than positive integers
10. If the function $f(x)$ and $g(x)$ are defined on $R \rightarrow R$ such that
- $$f(x) = \begin{cases} 0, & x \in \text{rational} \\ x, & x \in \text{irrational} \end{cases}; g(x) = \begin{cases} 0, & x \in \text{irrational} \\ x, & x \in \text{rational} \end{cases}$$
- then $(f - g)(x)$ is
- 1) one -one & onto
 2) neither one -one nor onto
 3) one -one but not onto
 4) onto but not one -one
11. If $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$, where the inverse trigonometric functions take only the principal values, then the incorrect option (s) is (are)
- 1) $\sin \beta < 0$
 2) $\cos \beta > 0$
 3) $\cos(\alpha + \beta) > 0$
 4) $\cos \alpha < 0$
12. A line in a three dimensional space makes an angle $\theta \left(0 < \theta \leq \frac{\pi}{2}\right)$ with both the x and y axes. Then the exhaustive set of all values of θ is:
- 1) $\left(0, \frac{\pi}{4}\right]$
 2) $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$
 3) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$
 4) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$
13. The lines $\vec{r} = (\hat{i} - \hat{j}) + l(2\hat{i} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j}) + m(\hat{i} + \hat{j} - \hat{k})$ (where l, m are real parameters)
- 1) do not intersect for any values of l and m
 2) intersect for all values of l and m
 3) intersect when $l=2$ and $m = \frac{1}{2}$
 4) intersect when $l=1$ and $m=2$



- 72



18. In a 3D space there are 2 intersecting lines one of which is axis and the other is generator line which rotates around the axis with respect to the point of intersection to form two napped cones with common apex. If α is the angle between the axis and the generator line and let a plane p be cutting two napped cones and makes an angle β with the axis ($\alpha < 90^\circ$) ($\beta \leq 90^\circ$) then choose the correct alternative(s) α is fixed

COLUMN-I (CONDITION)		COLUMN-2 (CROSS SECTION)	
I)	$\alpha = \beta$	P)	Circle
II)	$\beta = 90^\circ$	Q)	Ellipse
III)	$\alpha > \beta$	R)	Hyperbola
IV)	$\alpha < \beta$	S)	Parabola

- 1) I-S; II-P; III-R; IV-Q 2) I-S; II-P; III-Q; IV-R
 3) I-P; II-S; III-R; IV-Q 4) I-P; II-S; III-Q; IV-R
19. STATEMENT-1: A quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots then $p(p(x)) = 0$ has complex roots which are not purely imaginary
 STATEMENT-2: $K = \{\text{Arithmetic progression(s) with real terms ; } S_n \text{ as sum to } n \text{ terms, and } S_n = 0 \text{ has two non-zero real and distinct roots}\}$ then $6 \leq n(K) \leq 8$ (S_n denotes sum of the first 'n' terms with real terms)
 1) Statement-1 is True, statement-2 is True
 2) Statement-1 is False, Statement-2 is False
 3) Statement-1 is True, Statement-2 is False.
 4) Statement-1 is False, Statement-2 is True.
20. Let S be the standard deviation of n real observations. Each of the n observations is multiplied by a real constant c . Then the standard deviation of the resulting observations is
 1) s 2) cs 3) $-cs$ 4) cs or $-cs$





SECTION-II

(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 3), (1, 4)\}$ be a relation on A. Let S be the equivalence relation on A such that $R \subset S$ and the number of elements in S is n. Then the minimum value of n is _____
22. Consider $ax + by + c = 0$, $a, b, c \in R$. In XY – plane.
Number of incorrect statements among the following is/are
i) $a = 0, b = 0$ and $c = 0$ then the equation represents entire two-dimensional XY plane
ii) $a \neq 0, b = 0$ and $c \neq 0$ then the equation represents a line, parallel to X-axis
iii) $a = 0, b \neq 0$ and $c \neq 0$ then the equation represents a line, parallel to Y-axis
iv) $a \neq 0, b \neq 0$ and $c \neq 0$ then the equation represents a line, passing through origin
23. If three vertices of a n-sided regular polygon are joined to form a triangle then consider the following statements.
i) Number of obtuse angled triangles is equal to $(2k) \times {}^{k-1}C_2$ if $n = 2k + 1$
ii) Number of equilateral triangles is $\frac{n}{3}$ when n is multiple of 3.
iii) Number of right angled triangles = 0
iv) Number of isosceles triangles = $\frac{n(n-1)}{2} - 2k$, when $n = 3k$
Number of the correct statements among the above is/are
24. For a square matrix A with elements as real numbers, consider the following statements
i) Trace (AA^T) is nonnegative
ii) $Adj(kA) = k^{n-1} \times Adj(A)$, $k \in R$
iii) The maximum number of distinct elements is 780,
if A is triangular matrix of order 39.
iv) $(kA)^{-1} = \frac{1}{k} A^{-1}$, $k \in R - \{0\}$, $|A| \neq 0$
Number of correct statement(s) among the above is (are)
25. If $\sin(\alpha + \beta) + 2 \sin\left(\frac{2\alpha - \beta}{2}\right) \cos\left(\frac{\beta}{2}\right) = \sqrt{2} \cos \beta + \frac{1}{\sqrt{2}}$.
Let $p = \alpha + \beta$; $\alpha, \beta \in [0, 2\pi]$, also the maximum value of p is $\frac{m\pi}{n}$ where m, n are co-primes, then the value of $m - n$ is _____





PHYSICS

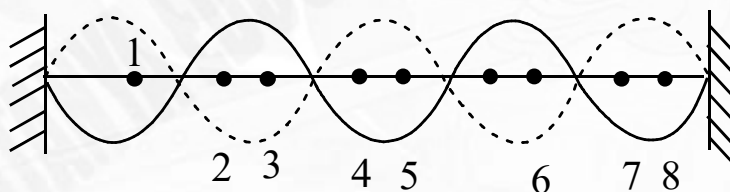
Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

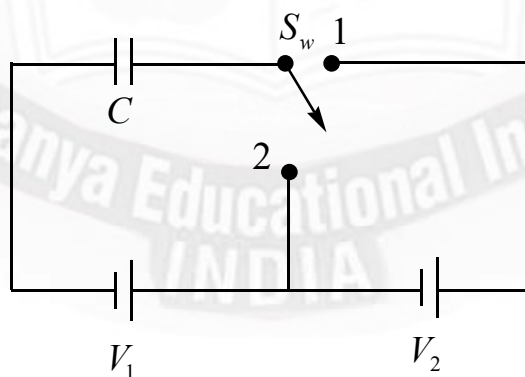
This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

26. A composite rod made up of two rods AB and BC are joined at B. The rods are of equal length at room temperature and have equal masses. The coefficient of linear expansion α of AB is more than that of BC. The composite rod is suspended horizontal by means of a thread at B. When the rod is heated:
- 1) It remains horizontal
 - 2) It tilts down on the side of AB
 - 3) It tilts down on the side of BC
 - 4) centre of mass of AB does not move
27. A stretched string is vibrating in its 5th harmonic as shown. Consider a particle 1 (figure). At an instant this particle is at mean positions and is moving towards its negative extreme. Which of the following set of particles, are in same plane with particle 1.



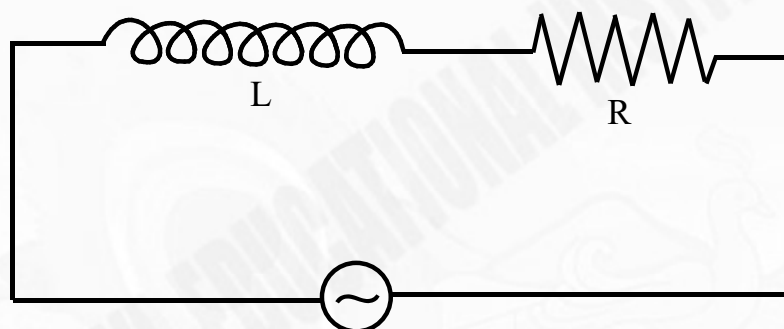
- 1) 2,4,7
 - 2) 3,5,6
 - 3) 5,7,8
 - 4) 2,4,6
28. The switch S_w is shifted from position 1 to position 2 as shown in the figure. Heat generated in the circuit is:





- 1) Independent of V_1 but dependent of V_2
- 2) Independent of V_2 as well as V_1
- 3) Dependent on V_1 but independent of V_2
- 4) Dependent on V_2 as well as V_1

29. In a simple L-R circuit with A.C. source at any instant the potential difference across inductor and resistance are V_L and V_R respectively and A.C. source has potential difference V_{AC} at the same instant. Then:



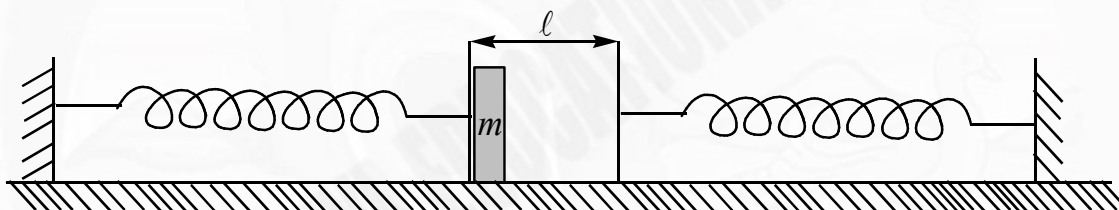
A.C. Source

- 1) $V_L^2 + V_R^2 = V_{AC}^2$ 2) $V_L + V_R = V_{AC}$ 3) $V_L^2 - V_R^2 = V_{AC}^2$ 4) $V_L^3 + V_R^3 = V_{AC}^3$
30. A particle is moving along x-axis back and forth in a box of length L. Assuming de-broglie hypothesis applicable for the particle, and it is moving with constant speed in the box making perfectly elastic collisions to the walls. The possible value of momentum of the particle is:
- (Note : h is plank's constant, n is principal quantum number)
- 1) $\frac{nh}{2L}$ 2) $\frac{nh}{L}$ 3) $\frac{nh}{4L}$ 4) $\frac{2nh}{3L}$
31. Backlash error corresponds to:
- 1) Change in circular scale reading without change in length between screw and stud
 - 2) Change in circular scale reading when second time same measurement is taken
 - 3) Change in main scale reading without moving screw head
 - 4) All of these





32. The astronauts aboard a space shuttle in circular orbit around the Earth wish to transfer the shuttle to a new orbit of large radius, the shuttle will have
- 1) a smaller orbital speed and a larger total energy
 - 2) a smaller orbital speed and a smaller total energy
 - 3) a larger orbital speed and a larger total energy
 - 4) a larger orbital speed and a smaller total energy.
33. A block of mass m is oscillating on smooth surface between two light springs of spring constant k separated by a distance ℓ colliding elastically with the springs. If the velocity of the block is increased by an external impulse when it is not touching either of the spring then time period

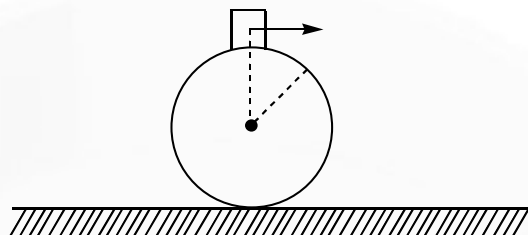


- 1) Increases.
 - 2) Decreases
 - 3) Remains same
 - 4) Time period is independent of ℓ
34. **Statement-1:** A rigid disc rolls without slipping on a fixed rough horizontal surface with uniform angular velocity. Then the acceleration of lowest point on the disc is zero.
- Statement-2:** For a rigid disc rolling without slipping on a fixed rough horizontal surface, the velocity of the lowest point on the disc is always zero.
- 1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 - 2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 - 3) Statement-1 is true, statement-2 is false.
 - 4) Statement-1 is false. Statement-2 is true.





35. An object is at the top of a smooth sphere which is kept fixed. As object slides down after being given a negligible side push, magnitude of acceleration of object during its motion till it reaches ground.



- 1) Increased and then decreases
 - 2) Decreases then becomes constant
 - 3) Increases and then becomes constant
 - 4) Continuously increases.
36. Current sensitivity of a moving coil galvanometer is increased by 50% and its resistance is increased so that new resistance is twice of the initial resistance. Then its new voltage sensitivity will be
- 1) half of the original value
 - 2) double of the original value
 - 3) $\frac{3}{2}$ times of the original value
 - 4) $\frac{3}{4}$ th of the original value
37. An series LCR circuit is resonating with a source whose emf varies with time as described in diagram-1. If we replace source by another source whose emf varies with time according to diagram-2, then:
(Assume both diagrams are on the same scale)

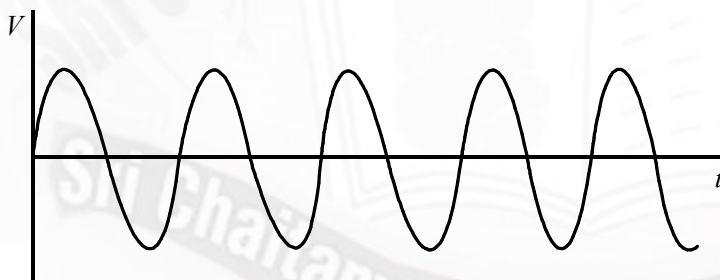


diagram 1

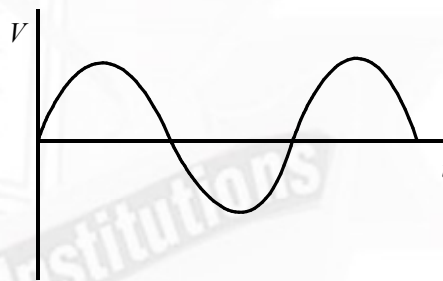


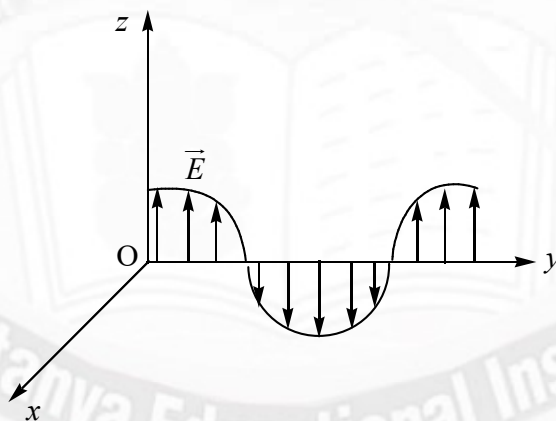
diagram 2

- 1) For getting resonance again, decrease R
- 2) Current will remain in phase with source voltage
- 3) For getting resonance again, decrease C
- 4) Current will lead source voltage after replacing the source





38. Suppose a small cylinder of a paramagnetic substance and a small cylinder of a diamagnetic substance are placed between the (parallel) pole pieces of a strong magnet, so that they are aligned along the direction of the field lines. Then they will
- 1) Attract when they are placed such that their axis is common and repel when they are placed side by side.
 - 2) Repel when they are placed such that their axis is common and attract when they are placed side by side.
 - 3) Attract when they are placed such that their axis is common as well as when they are placed side by side.
 - 4) Repel when they are placed such that their axis is common as well as when they are placed side by side.
39. An unknown nucleus collides with a ${}^4\text{He}$ nucleus, and after the collision the two nuclei travel in perpendicular directions relative to each other. If kinetic energy is lost in the collision, the unknown nucleus must be:
- 1) ${}^{28}\text{N}$
 - 2) ${}^4\text{He}$
 - 3) ${}^{12}\text{C}$
 - 4) a nucleus with mass lighter than ${}^4\text{He}$
40. The electric field for a plane electro-magnetic wave travelling in +y-direction is shown in the figure. Consider a point where electric field is in the +z direction. The magnetic field $\vec{B}$ is

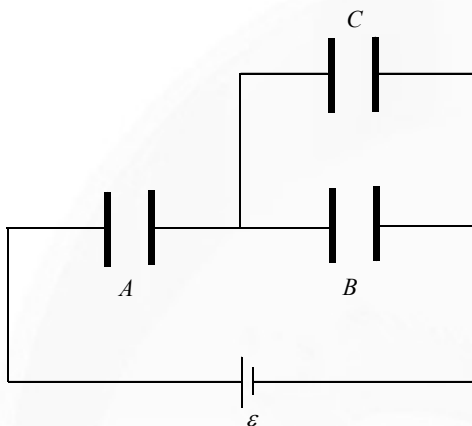


- 1) In the +x direction and in phase with the electric field $\vec{E}$
- 2) In the -x direction and in phase with the electric field $\vec{E}$
- 3) In the +z direction and in phase with the electric field $\vec{E}$
- 4) In the -z direction and in phase with the electric field $\vec{E}$

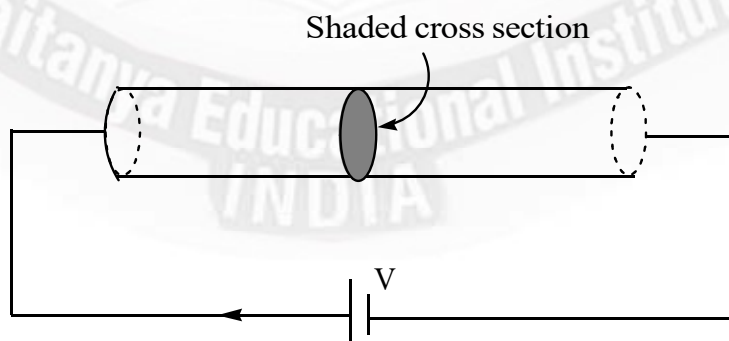




41. Three capacitors A, B and C are connected with battery of emf ε . All capacitors are identical initially. If dielectric slab is inserted between plates of capacitor A slowly with help of external force then



- 1) Chemical energy of battery remains unchanged
 - 2) Positive work is done on the given system by external force.
 - 3) Force between plates of capacitor C decreases.
 - 4) Magnitude of change of charge on B and C is same
42. Column-1 gives physical quantities based on a situation in which an ideal cell of emf V is connected across a cylindrical rod of uniform cross-section area and conductivity (σ) as shown in figure. E, J, ϕ and i are electric field at, current density through, electric flux through shaded cross-section respectively as shown in figure. Physical quantities in Column-II are related to those in Column-I. Match the expression in Column-I with the statements in Column-II.





	List-I		List-II
A	$\frac{\phi}{i}$	P	Conductivity of rod
B	$\frac{E}{J}$	Q	Resistance of rod
C	$\sigma\phi V$	R	Resistivity of rod
D	$\frac{V}{\sigma\phi}$	S	Power delivered to rod

- 1) A – R, B – R, C – S, D – Q 2) A – P, B – R, C – S, D – Q
 3) A – R, B – R, C – P, D – Q 4) A – R, B – Q, C – S, D – R

43. **Assertion (A):** Under electrostatic condition, the electric field at surface of conductor can have no component parallel to surface.

Reason (R): The surface of conductor is equipotential and no work is done by electric field to move the charge on surface.

- 1) A and R are True and R is a correct explanation for A
 2) A and R are True but R is not a correct explanation for A
 3) A is True, R is False
 4) A is False, R is True

44. **Assertion (A):** A block released from top of a rough inclined plane will reach bottom with the same kinetic energy whatever be the angle of inclination and height as long as the horizontal displacement remains same.

Reason (R): In a rough inclined plane situation, for a given horizontal dimension both the magnitudes of friction force and displacement change as angle of inclination changes.

- 1) A and R are True and R is a correct explanation for A
 2) A and R are True but R is not a correct explanation for A
 3) A is True, R is False
 4) A is False, R is True





45. **Assertion (A):** The ratio of molar specific heats $\frac{C_p}{C_v}$ for He is more than H_2 .

Reason (R): Mass of He atom is more than that of H_2 .

- 1) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion
- 2) Both Assertion & Reason are incorrect
- 3) Assertion is correct but Reason is incorrect
- 4) Assertion is incorrect but Reason is correct

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

Note:

From: Question number's 46 to 50 rules for Answer marking

If your answer is options 2,4 Then you have to fill the OMR sheet as '24'

If your answer is options 1,3,4 Then you have to fill the OMR sheet as '134'

And if your answer is options 1,2,3,4 then you have to fill the OMR sheet as '1234'

46. Pick the CORRECT statement(s) from the following:
- 1) Viscosity in liquids, is the property which is responsible for transport of momentum from one layer of liquid to other.
 - 2) Viscous resistance force in fluids and kinetic friction force in solids are non-conservative forces.
 - 3) Ideally, a solid steel sphere, starting from rest, requires infinite time to acquire terminal velocity in a big reservoir of honey.
 - 4) In liquids, the cause of viscosity, is the cohesive force.
47. A bar magnet is moved along the axis of copper ring placed far away from the magnet. Looking from the side of the magnet, an anticlockwise current is found to be induced in the ring. Which of the following may be true?
- 1) The south pole faces the ring and the magnet moves towards it.
 - 2) The north pole faces the ring and the magnet moves towards it.
 - 3) The south pole faces the ring and the magnet moves away from it.
 - 4) The north pole faces the ring and the magnet moves away from it.

Sec: **Sr. XL-500, Sr.S60(N&S) & LIIT**

Page 17



Sri Chaitanya
Educational Institutions



**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

**JEE MAIN
2023**

**SINGARAJU
VENKAT KORDINNYA**

**SR. CHAITANYA
EDUCATIONAL INSTITUTIONS**

ALL-INDIA RANK 1

RANK

1

**JEE Advanced
2023**

**VAVILALA
CHANDRASEKHAR REDDY**

**SR. CHAITANYA
EDUCATIONAL INSTITUTIONS**

ALL-INDIA RANK 1

RANK

1

**NEET
2023**

BORA VARUN

**SR. CHAITANYA
EDUCATIONAL INSTITUTIONS**

ALL-INDIA RANK 1

RANK

1

For More Join: @ITTJEE_Advanced



48. Which of the following statement(s) is/are correct?

- 1) If a moving charged particle enters into a region of uniform magnetic field from outside, it does not complete a circular path.
- 2) If a moving charged particle traces a helical path in a uniform magnetic field, the axis of the helix is parallel to the magnetic field.
- 3) The power associated with the force exerted by a magnetic field on a moving charged particle is always equal to zero.
- 4) If in a region a uniform magnetic field and a uniform electric field both exist, a charged particle moving in this region cannot trace a circular path

49. A lens of focal length 'f' is placed in between an object and screen at a distance 'D'. The lens forms two real images of object on the screen for two of its different positions, a distance 'x' apart. The two real images have magnifications m_1 and m_2 respectively ($m_1 > m_2$) Then:

- 1) $f = \frac{x}{m_1 - m_2}$ 2) $m_1 m_2 = 1$ 3) $f = \frac{D^2 - x^2}{4D}$ 4) $D \geq 4f$

50. A and B are input of the logic gate implementations and Y is the output. Which of the following option(s) implement the output $Y = AB + \overline{A}\overline{B}$.

- 1)
- 2)
- 3)
- 4)

Sec: Sr. XL-500, Sr.S60(N&S) & BLIIT

Page 18



Sri Chaitanya
Educational Institutions

Infinity
Learn



THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023

JEE MAIN
2023

SINGARAJU
VENKAT KORDINYA
RANK 1



RANK
1

JEE Advanced
2023

VAVILALA
CHIVILAS REDDY
RANK 1



RANK
1

NEET
2023

BORA VARUN
CHAKRAVARTHI
RANK 1



RANK
1

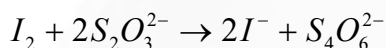
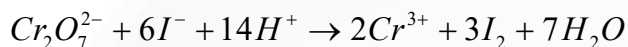
For More Join: @ITTJEE_Advanced

**CHEMISTRY****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. In an iodometric titration, a solution of $Na_2S_2O_3$ is standardized against a known mass of pure $K_2Cr_2O_7$. The reactions involved are:



Which of the following statements are correct?

A) The equivalent weight of $K_2Cr_2O_7$ in this process is $\frac{M}{6}$.

B) The n -factor for I_2 in the second reaction is 2.

C) The total number of equivalents of $K_2Cr_2O_7$ used must equal the total number of equivalents of $Na_2S_2O_3$ consumed.

D) $Na_2S_2O_3$ acts as a reducing agent in the second step.

1) A,C,D 2) B,D 3) A,C 4) A,B,C,D

52. For the hydrogen atom, which of the following statements are correct?

A) The number of radial nodes in the 3p orbital is 1

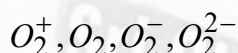
B) The 2s and 2p orbitals have identical energy in H but different radial distribution

C) The probability of finding the electron at the nucleus in a p-orbital is zero

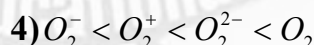
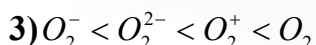
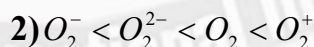
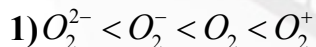
D) The 4s orbital has three radial nodes

1) B,C 2) A,B,C 3) C,D 4) A,B,C,D

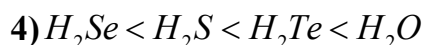
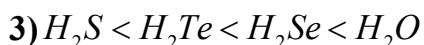
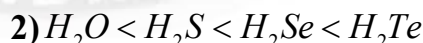
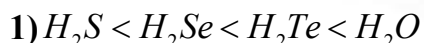
53. Consider the ions/molecule



For increasing bond order the correct option is:



54. The correct order of melting points of hydrides of group 16 elements is:



Sec: **Sr. XL-500, Sr.S60(N&S) & BLIIT**

Page 19



Sri Chaitanya
Educational Institutions



**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

**JEE MAIN
2023**

SINGARAJU
VENKAT KORDINNYA
RANK 1

**RANK
1**

**JEE Advanced
2023**

VAVILALA
CHIVILAS REDDY
RANK 1

**RANK
1**

**NEET
2023**

BORA VARUN
CHAKRAVARTI
RANK 1

**RANK
1**

For More Join: @ITTJEE_Advanced



55. Given below are two statements.

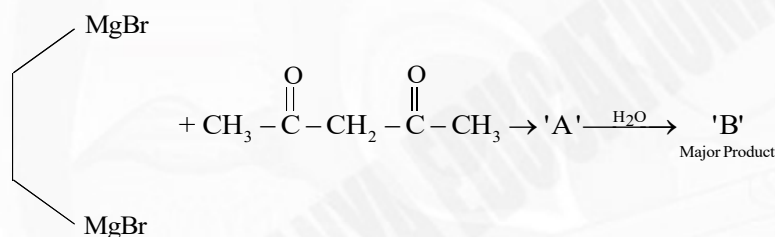
Statement I: In 'Lassaigne's Test', when both nitrogen and sulphur are present in an organic compound, sodium thiocyanate is formed.

Statement II: If both nitrogen and sulphur are present in an organic compound, then the excess of sodium used in sodium fusion will decompose the sodium thiocyanate formed to give NaCN and Na_2S .

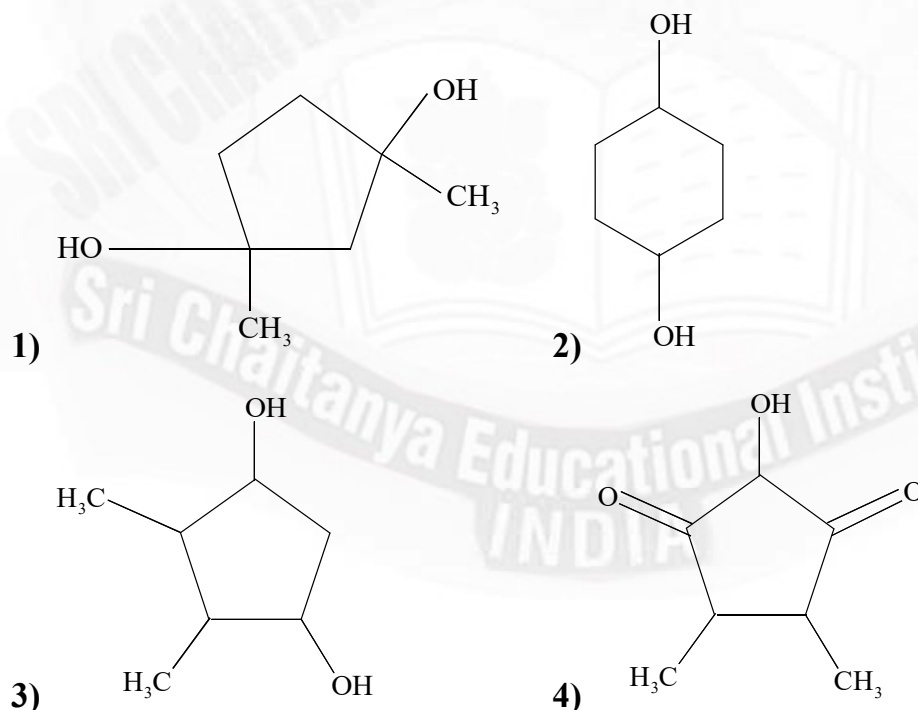
In the light of the above statements, choose the **most appropriate** answer from the options given below:

- 1) Both **Statement I** and **Statement II** are correct.
- 2) Both **Statement I** and **Statement II** are incorrect.
- 3) **Statement I** is correct but **Statement II** is incorrect.
- 4) **Statement I** is incorrect but **Statement II** is correct.

56.

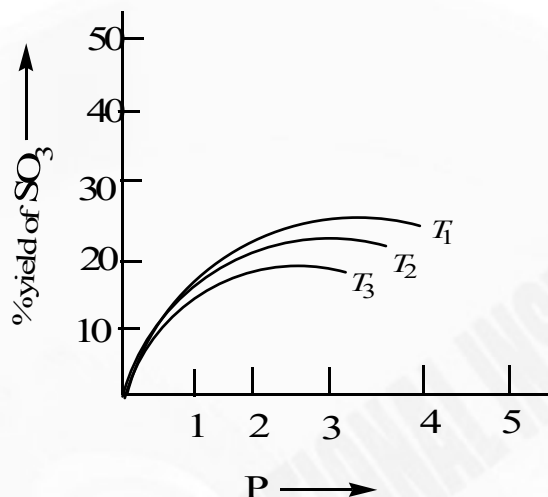
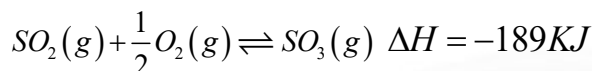


Consider the above reaction sequence and identify the product B.





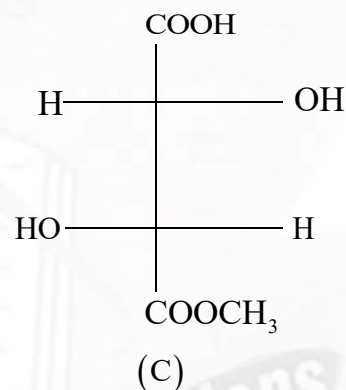
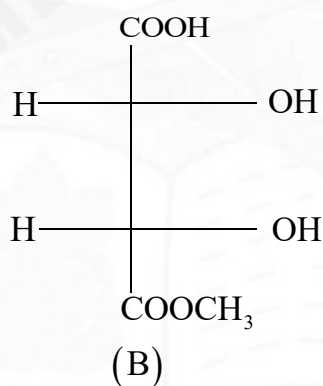
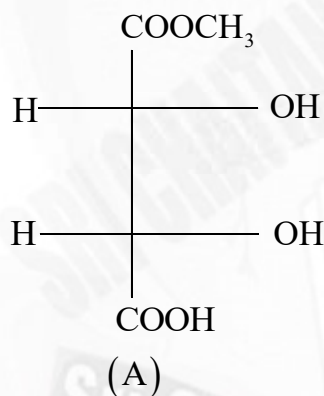
57. The preparation of $SO_3(g)$ by the reaction,



Show the following figure in % yield against pressure at different temperature, then:

- 1) $T_3 > T_2 > T_1$ 2) $T_1 > T_2 > T_3$ 3) $T_1 = T_2 = T_3$ 4) $T_2 > T_1 > T_3$

58. The correct statement about the compound A, B and C is



- 1) 'A' and 'B' are identical 2) 'A' and 'B' are diastereomers
3) 'A' and 'C' are enantiomers 4) 'A' and 'B' are enantiomers

59. Observe the following molecules $PCl_5, BrF_5, ClF_5, PF_5, ClF_3, XeF_4, XeF_2, IF_5$. The number of molecules having square pyramidal geometry from the above is:

- 1) 4 2) 5 3) 3 4) 6





60. Which is the correct match for all?

LIST-I		LIST-II	
A	Sucrose	P	$\beta - (1,4)$ – Glycosidic linkage
B	Maltose	Q	$\alpha, \beta - (1,2)$ – Glycosidic linkage
C	Starch	R	natural biopolymer
D	Cellulose	S	Silver mirror with tollen's reagent

1) A-S; B-P; C-R; D-Q

2) A-Q; B-S; C-P; D-R

3) A-Q; B-S; C-R; D-P

4) A-R; B-S; C-Q; D-P

61. Which one of the following statements is correct?

1) Two different solutions of urea of same molality will have same depression in freezing point.

2) Best method to calculate molar mass of biomolecules is via relative lowering in vapour pressure.

3) Decreasing order of osmotic pressure at the same temperature for 0.1 M aqueous solution of $BaCl_2$, KCl , and sucrose is $BaCl_2 > KCl > sucrose$

4) Van't Hoff factor of 0.1 M $NaCl$ and 0.1 M $CaCl_2$ aq. solution is same

62. Which of the following pair of acid radicals cannot be distinguished using dil. H_2SO_4 ?

1) S^{2-} , NO_3^-

2) Cl^- , Br^-

3) CO_3^{2-} , Br^-

4) CO_3^{2-} , I^-

63. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Synthesis of ethyl phenyl ether may be achieved by Williamson synthesis.

Reason (R) : Reaction of bromobenzene with sodium ethoxide yields ethyl phenyl ether.

In the light of the above statement, choose the most appropriate answer from the options given below:

Sec: Sr. XL-500, Sr.S60(N&S) & LIIT

Page 22



Sri Chaitanya
Educational Institutions

Infinity
Learn



**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

JEE MAIN
2023

SINGARAJU
VENKAT KORDINNYA
RANK 1

RANK
1

JEE Advanced
2023

VAVILALA
CHIVILAS REDDY
RANK 1

RANK
1

NEET
2023

BORA VARUN
CHAKRAVARTI
RANK 1

RANK
1

For More Join: @ITTJEE_Advanced



- 1) (A) is not correct but (R) is correct
 2) (A) is correct but (R) is not correct
 3) Both (A) and (R) correct but (R) is NOT the correct explanation of (A)
 4) Both (A) and (R) are correct and (R) is the correct explanation of (A)
64. Which of the following electronic configuration of the corresponding element is wrongly matched?
 1) Thulium – $[Xe]4f^{13}6s^2$ 2) Americium – $[Rn]5f^77s^2$
 3) Uranium – $[Rn]5f^77s^2$ 4) Terbium – $[Xe]4f^96s^2$
65. Which of the following statements are correct about corrosion?
 (i) Corrosion of Fe can be prevented by coating with bisphenol
 (ii) Green coating on bronze is due to corrosion of Zn
 (iii) Corrosion is an electrochemical phenomenon
 (iv) By providing a sacrificial electrode like Mg or Zn, corrosion of iron can be prevented
 1) (i), (ii) and (iii) only 2) (ii) and (iv) only
 3) (iii) and (iv) only 4) (i), (iii) and (iv) only
66. Assertion: For each $10^\circ C$ rise of temperature the k is nearly double or triple.
 Reason: Energy wise distribution of molecules in a gas sample is an exponential function of temperature $e^{-E_a/RT}$ so is doubled or tripled.
 1) Both 'A' and 'R' are true and 'R' is the correct explanation of 'A'.
 2) Both 'A' and 'R' are true and 'R' is not the correct explanation of 'A'.
 3) 'A' is true and 'R' is false
 4) Both 'A' and 'R' are false
67. BCl_3 easily accepts a lone pair of electrons from ammonia to form $BCl_3 \cdot NH_3$ Then identify the incorrect statement for the given reaction
 1) Hybridisation of Boron is same in Both BCl_3 and $BCl_3 \cdot NH_3$
 2) Hybridisation of nitrogen is same in both NH_3 and $BCl_3 \cdot NH_3$
 3) Trigonal planar Boron changes to tetrahedral boron
 4) There is no change in the electron pair structure around nitrogen
68. Which of the following mixtures can act as a buffer?
 1) $NaOH + HCOONa$ (1 : 1 molar ratio) 2) $HCOOH + NaOH$ (2 : 1 molar ratio)
 3) $NH_4Cl + NaOH$ (2 : 1 molar ratio) 4) $HCOOH + NaOH$ (1 : 1 molar ratio)





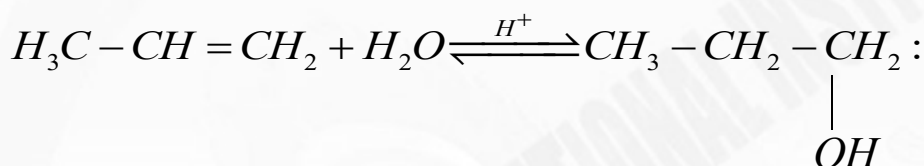
69. Given below are two statements:

Statement-I: The radius of 5d series element is nearly similar to the 4d series element.

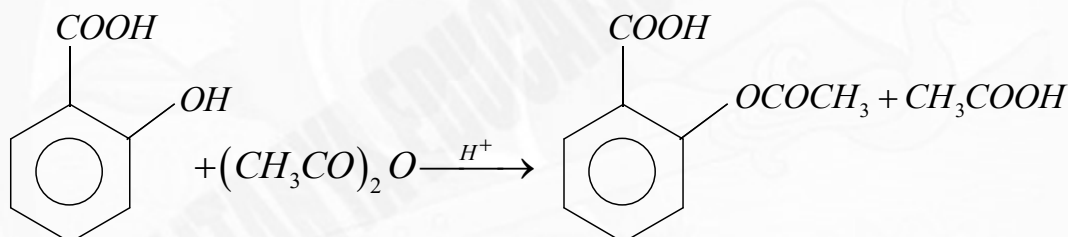
Statement-II: Due to poor shielding effect of 4f e⁻ Effective nuclear charge of 5d series elements is greater than that of 4d series elements. In the light of the above statements, choose the most appropriate answer from the options given below:

- 1) Statement I & II both are correct.
- 2) Statement I is correct but II are incorrect.
- 3) Statement I is incorrect but II is correct.
- 4) Both statement I & II are incorrect.

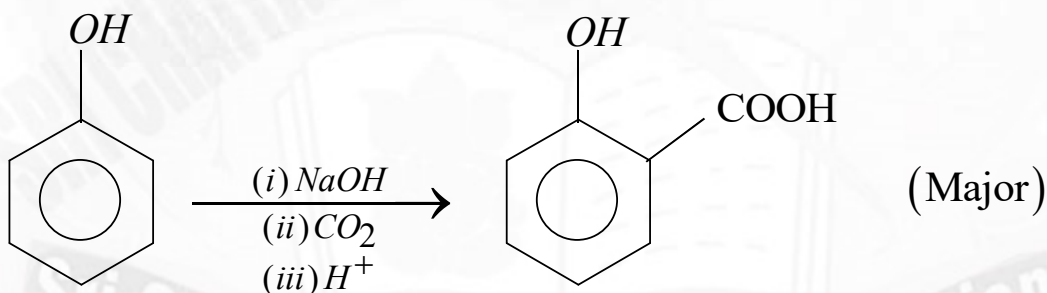
70. Choose the correct pair of reactions among the given below reactions.



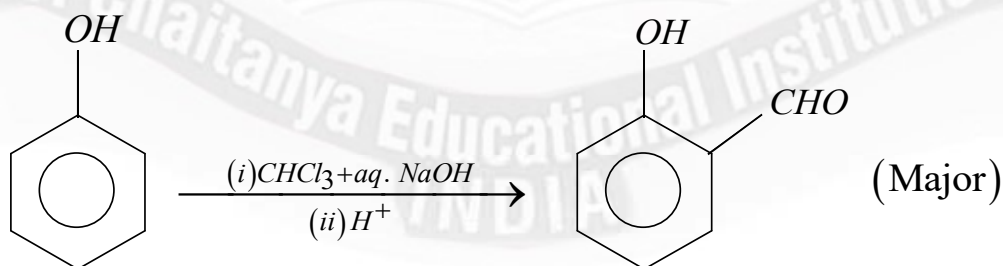
a)



b)



c)



d)

1) a, b

2) a, b, c

3) b, c, d

4) a, b, c, d





SECTION-II

(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. The spin-only magnetic moment value of an octahedral complex among $\text{CoCl}_3 \cdot 4\text{NH}_3$, $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{PtCl}_4 \cdot 2\text{HCl}$, which upon reaction with excess of AgNO_3 gives 2 moles of AgCl is _____ B.M. (Nearest Integer)
72. Aniline and N-methylaniline can be distinguished by using, how many of the following tests?
- (I) Tollen's reagent (II) Hinsberg reagent (III) Carbylamine test
(IV) Mulliken Test (V) Azo dye test (VI) Bayer's test
(VII) Lucas test (VIII) Hofmann mustard oil test
73. How many of the following are intensive properties
- i) Boiling Point ii) Volume iii) Specific Gravity iv) Heat Capacity
v) Density vi) Enthalpy vii) Viscosity viii) Mass
ix) Specific heat capacity x) Gibbs Free Energy
74. The number of aromatic compounds among the following is X. What is X?
- (i) Cyclopentadienyl anion (ii) Tropylium cation
(iii) Cyclopropenyl cation (iv) Cyclo octatetraene
75. A chloro compound "A",
- i) Forms aldehydes on ozonolysis followed by the hydrolysis.
ii) When vaporized completely 1.81g of A, gives 448 ml of vapour at STP. (22.4 lits)
- The number of carbon atoms in a molecule of compounds A is _____





Sri Chaitanya
Educational Institutions

Infinity
Learn



Sri Chaitanya
Techno School
The right mentor for IIT (JEE), NEET, Olympiad & all Other Competitive exams



JEE MAIN 2025

31 STUDENTS BELOW 100 AIR

1

ALL INDIA OPEN CATEGORY RANK

300
300

VANGALA AJAY REDDY
APP. NO. 250302250592
CLASSROOM STUDENT FROM GRADE IX - XI

1

ALL INDIA OPEN CATEGORY RANK

300
300

DEVPUTTA MAJHI
APP. NO. 25030306185
DLP/AITS STUDENT

9

ALL INDIA OPEN CATEGORY RANK

295
300

TOSHNIWAL SHIVEN
APP. NO. 250303091420
DLP/AITS STUDENT

10

ALL INDIA OPEN CATEGORY RANK

295
300

SAKSHAM JINDAL
APP. NO. 25030236696
DLP/AITS STUDENT

BELOW
100
ALL INDIA OPEN
CATEGORY RANKS

31

BELOW
500
ALL INDIA OPEN
CATEGORY RANKS

95

BELOW
10
ALL INDIA CATEGORY
RANKS COUNT

10

BELOW
100
ALL INDIA CATEGORY
RANKS COUNT

98

BELOW
1000
ALL INDIA CATEGORY
RANKS COUNT

579

TOTAL QUALIFIED RANKS
FOR JEE ADVANCED-2025

22,094

*DLP/AITS

JEE 2025 STARS SHINE BRIGHT

Sri Chaitanya Tops JEE ADVANCED

ALL INDIA OPEN CATEGORY RANKS

AIR

1

RUTVIK SAI
H.T.No. 256055278 (OBC-NCL)

AIR

3

MAJID MUJAHID HUSAIN
H.T.No. 251134112*

AIR

5

UJJWAL KESARI
H.T.No. 252016104*

AIR

6

AKSHAT KUMAR CHAURASIA
H.T.No. 254065055*

BELOW
100
ALL INDIA OPEN
CATEGORY RANKS

29

BELOW
500
ALL INDIA OPEN
CATEGORY RANKS

113

BELOW
1000
ALL INDIA OPEN
CATEGORY RANKS

205

BELOW
1000
ALL INDIA CATEGORY
RANKS COUNT

745

NUMBER OF
QUALIFIED RANKS

4,212

*DLP/AITS



www.srichaitanya.net



0-66 06 06 06

For More Join: @IITJEE_Advanced